David Zambrano

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EDUCATION

Florida Polytechnic University, Lakeland, FL

Bachelor of Science in Computer Engineering – *GPA: 3.88* May 2026

Master of Science in Electrical Engineering – *GPA: 3.5* Graduating May 2027

PROFESSIONAL EXPERIENCE

Undergraduate & Graduate Research Assistant | *PolySilicon* May 2025 – Present

- Conducting thesis research in silicon photonic integrated circuit (PIC) design, from simulation through GDSII layout and tapeout on 220 nm Silicon-on-Insulator (SOI) process.
- Completed three end-to-end tapeouts with Applied Nanotools' NanoSOI MPW service (Oct 2025, Jun 2026, Oct 2026) including a Mach-Zehnder Interferometer, novel optical ring resonator, and all-optical logic gates (NOT, NAND, NOR), operating at the center of the optical C-band range (1550nm).
- Utilized Flexcompute's Tidy3D & Lumerical suite for device geometry specification and electromagnetic simulation, and KLayout, gdsfactory, and Luceda IPKISS for fabrication layout.

Electrical Systems Lead | *Phoenix Autonomy* Jan 2026 – Present

- Leading a six-person team within the Phoenix Autonomy research project, designing the autonomous boat's electrical system for the RoboBoat competition.
- Designed a thruster control scheme from scratch using an AM32 ESC & Raspberry Pi Pico (C++) for manual control (with FlySky controller & receiver) and autonomous control with command inputs from a Jetson Orin Nano.
- Conducted KiCad design workshops with undergrad team to design the boat's main board, complete with buck converters, an RP2040 MCU, and optocoupler - MOSFET - contactor circuit for high power switching to thrusters.

Embedded R&D Intern | *Measutronics Corporation* May 2024 – Aug 2024

- Developed a PLC-based alarm system to warn dredging vehicle operators of hazardous or restricted zones outside their line of sight.
- Created custom firmware (C++) for a robust & configurable system, complete with Windows Forms GUI (C#).

DESIGN PROJECTS

Subwavelength-engineered Moiré Ring Resonator | *PolySilicon* Aug 2025 – Present

- Designed a ring resonator with width-modulated sidewalls to induce azimuthal perturbation enabling tunable spectra, mode splitting, and localized resonances for filtering, nonlinear optics, and optical computing.
- Electromagnetic simulations demonstrate unique disorder-induced signatures and resonances with up to 26 dB extinction ratio, suitable for hardware security applications in the way of a Physical Unclonable Function (PUF).

Cascadable All-Optical Logic Gates | *PolySilicon* May 2025 – Present

- Designed an all-optical inverter from microring resonators, utilizing destructive interference effects for logic inversion for a passive configuration without the use of electro-optic or thermo-optic tuning.
- Electromagnetic simulations confirm transmission inversion, from which 2-input NAND and NOR gates are approximated. SEM imaging confirms high-fidelity geometry thanks to a relaxed passive design.

Custom 3D Printed Nerf Blaster "Warthog" | *Baja Blasters (Personal)* Dec 2025 – Aug 2026

- Designed a 4-layer printed circuit board in KiCad, incorporating an RP2040 to drive an electronic speed controller for two brushless motors, with a fast decay circuit for a solenoid pusher for fire rates up to 25 darts per second.
- Designed Buck-LDO circuit to supply 3.3V to low power circuitry, wrote firmware (C++) for reconfigurable user profile settings, memory management, and an I2C OLED screen for user interaction and system feedback.

RISC-V 32I Processor | *Florida Polytechnic University* Mar 2025 – May 2025

- Constructed a single-core RV32I processor subset in the Intel Quartus Prime FPGA SDK, implementing 23 arithmetic and branch logic instructions.
- Validated on DE-10 Lite FPGA board through purpose written assembly program to recursively calculate the Fibonacci sequence for 16 iterations.

PUBLICATIONS & PRESENTATIONS

- D. Zambrano and F. B. Tarik, "Design and Fabrication of a Subwavelength-engineered Silicon Photonic Moiré Ring Resonator," in CLEO 2026, Technical Digest Series (Optica Publishing Group, 2026), paper JW1.47. (20 May 2026); DOI: https://doi.org/10.1364/CLEO_AT.2026.JW1.47
- D. Brokaw, D. Zambrano, F. B. Tarik "Scalable all-optical logic gates using cascadable silicon photonic resonator inverters", Proc. SPIE 13900, Integrated Optics: Devices, Materials, and Technologies XXX, 139000E (4 Mar 2026); DOI: https://doi.org/10.1117/12.3081538
- F. Gonzalez, D. Zambrano, F. B. Tarik "Deeply sub-diffractive silicon photonic ring resonator for localized field enhancement in integrated photonics" *SPIE Photonics West*, 2026
- D. Zambrano and F. B. Tarik “Cascadable All-Optical Logic Gates in Silicon Photonics for Energy-efficient Computing and AI Acceleration” Oral Presentation, Second Annual Student Research Symposium IEEE Student Branch, Barnett Applied Research Center, (18 April 2026). (First Place Paper Presentation, Florida Polytechnic University)
- D. Zambrano and F. B. Tarik “Truly Azimuthal Superperiodic Silicon Photonic Microring Resonator for Scalable Post-Quantum Hardware Security” Poster Presentation, Second Annual Research Day, Barnett Applied Research Center, (27 April 2026). (First Place Outstanding Research Presentation, Florida Polytechnic University)

SKILLS

Simulation & Layout CAD: Lumerical (FDTD, MODE, INTERCONNECT), Flexcompute Tidy3D, Luceda IPKISS, Mentor Graphics Pyxis, KLayout, gdsfactory, MATLAB, Electric.

EDA & Embedded Tools: KiCad, STM32CubeIDE, Microchip Studio, Intel Quartus Prime, Multisim, LTspice, Visual Studio Code, SolidWorks (CSWA), Autodesk Fusion.

Programming & HDL: C, C#, C++, Python, Embedded C, MIPS assembly, Verilog & SystemVerilog.

Relevant Coursework: Optoelectronics, VLSI Design, Embedded Operating Systems, Computer Architecture and Organization, Microprocessors, Object Oriented Programming, Data Structures and Algorithms, Digital Logic Design.

LEADERSHIP EXPERIENCE & AWARDS

Leadership: Phoenix Autonomy Electrical Systems Lead (Jan 2026 – Present), Phoenix Autonomy Battery Lead (Aug 2025 – Dec 2025), Poly Concert Band Captain (Aug 2026 – Present), Poly Concert Band Vice Captain (Aug 2024 – May 2026), Violet Voices Choir Bass Section Lead (Aug 2024 – May 2025), Presidential Ambassador (Jan 2023 – May 2025).

Awards: First Place Research Presentation – Second Annual Poly Research Day, First Place Paper Presentation – Second Annual Student Research Symposium IEEE Student Branch, IEEE-HKN Honor Society Member, President’s & Provost’s List.